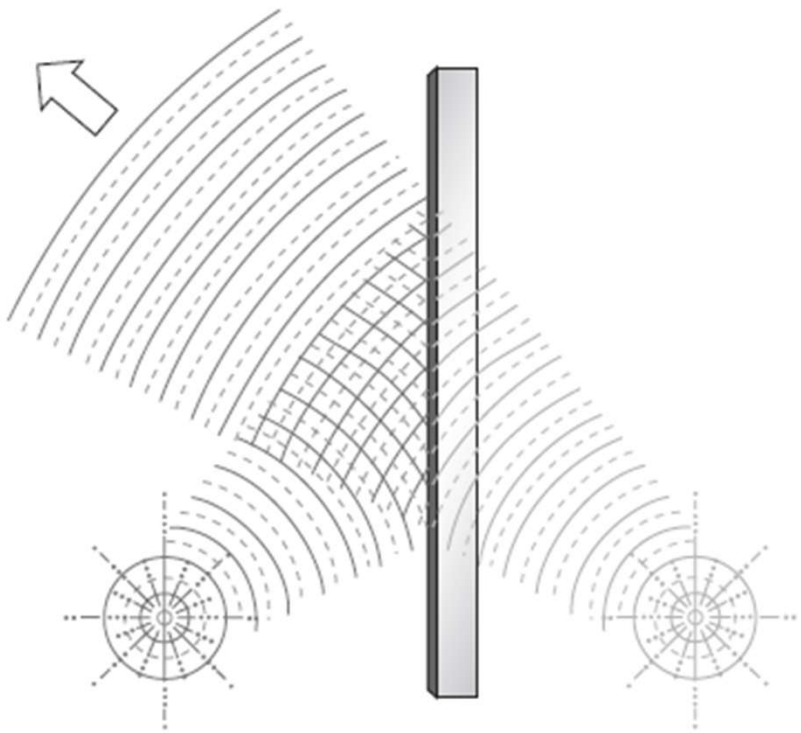
Óptica Geométrica

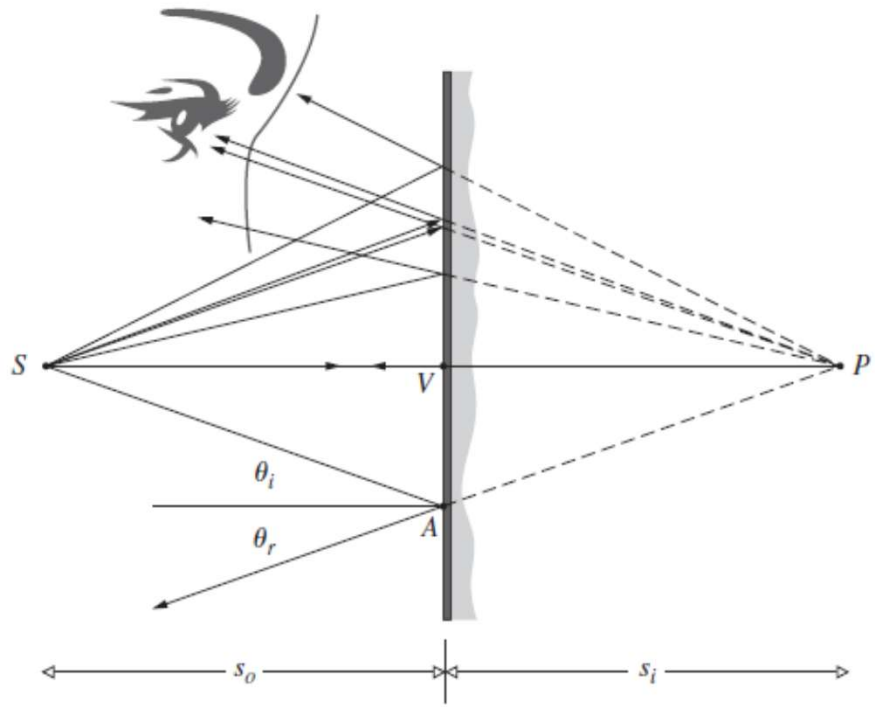
Física General IV / Física IV

Prof. Erick Cubillo

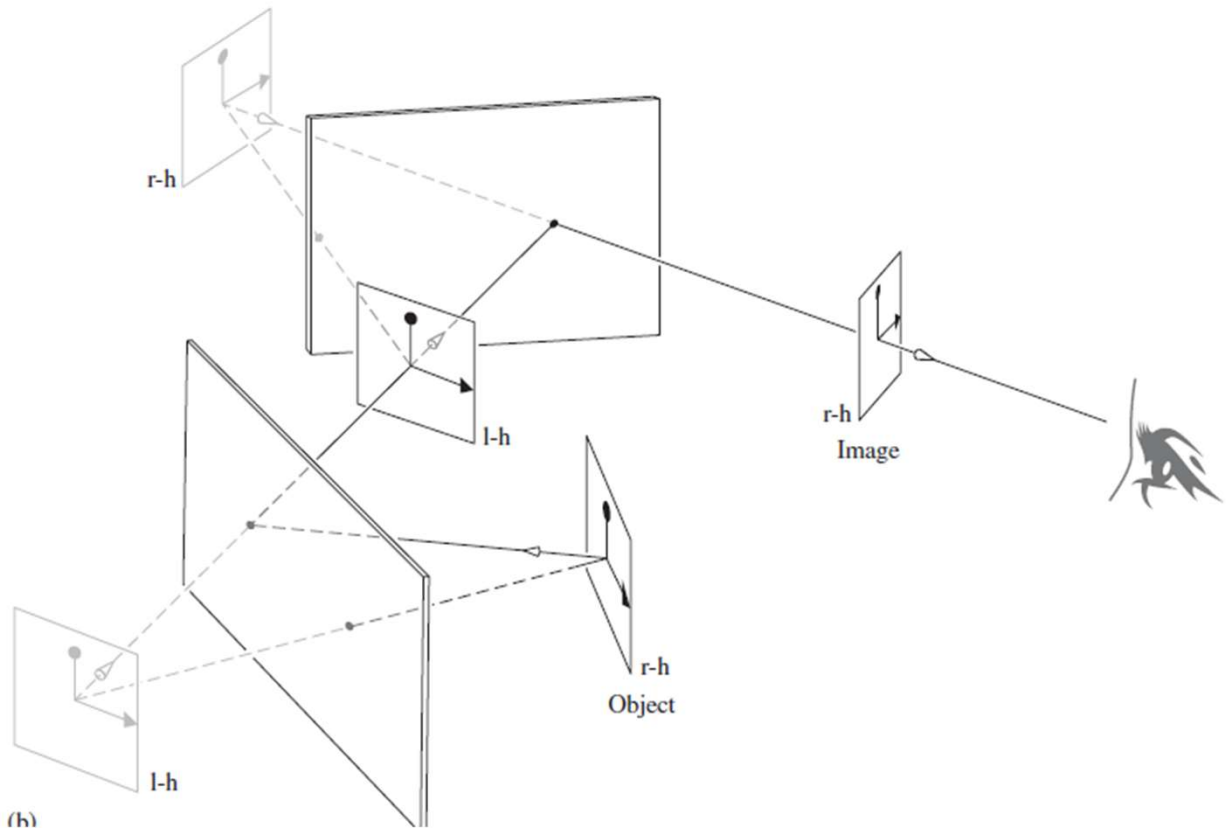
III 2025

Espejos planos



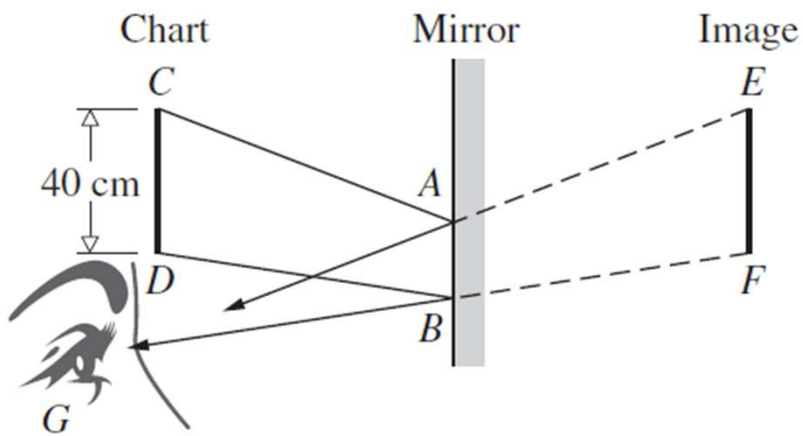


Espejos planos

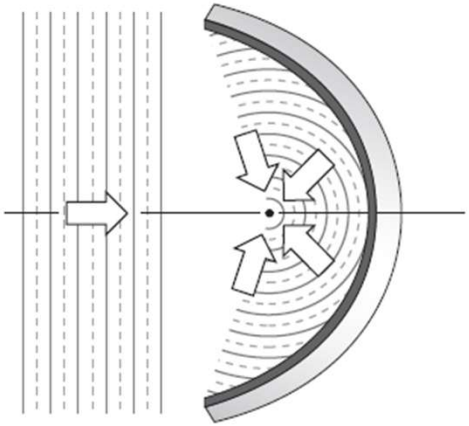


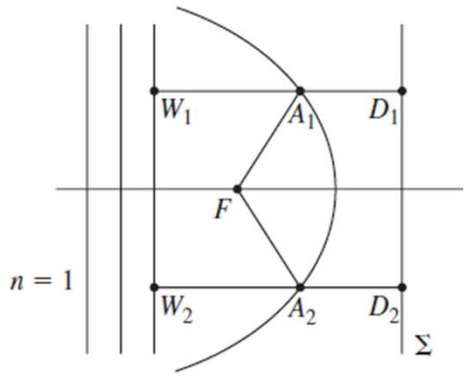
Espejos planos

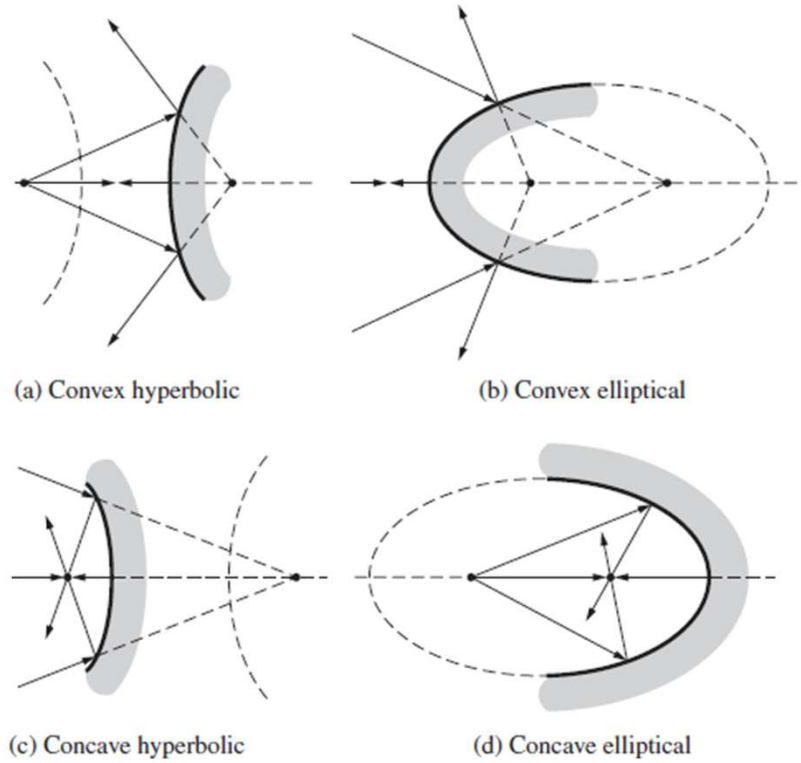
En un examen visual, se coloca un cartel de 40 x 20 cm sobre la cabeza de la persona. ¿Cuáles son las medidas mínimas del espejo para que el objeto completo se vea?



Espejos asféricos







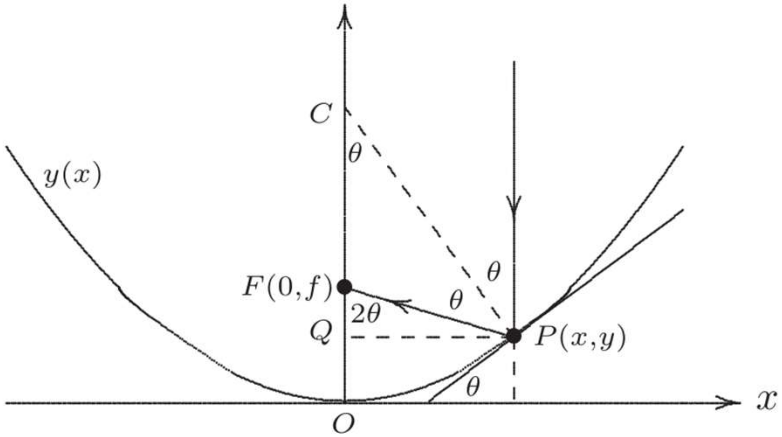
Espejos asféricos

Considere un espejo curvo cuya proyección en el plano $x - y$ está dada por $y(x)$ tal que es simétrico respecto al eje y.

a) Considerando un rayo de luz paralelo al eje de simetría que impacta en el punto $P(x, y)$ de la curva con un ángulo de incidencia θ, muestre que la distancia focal f está dada por:

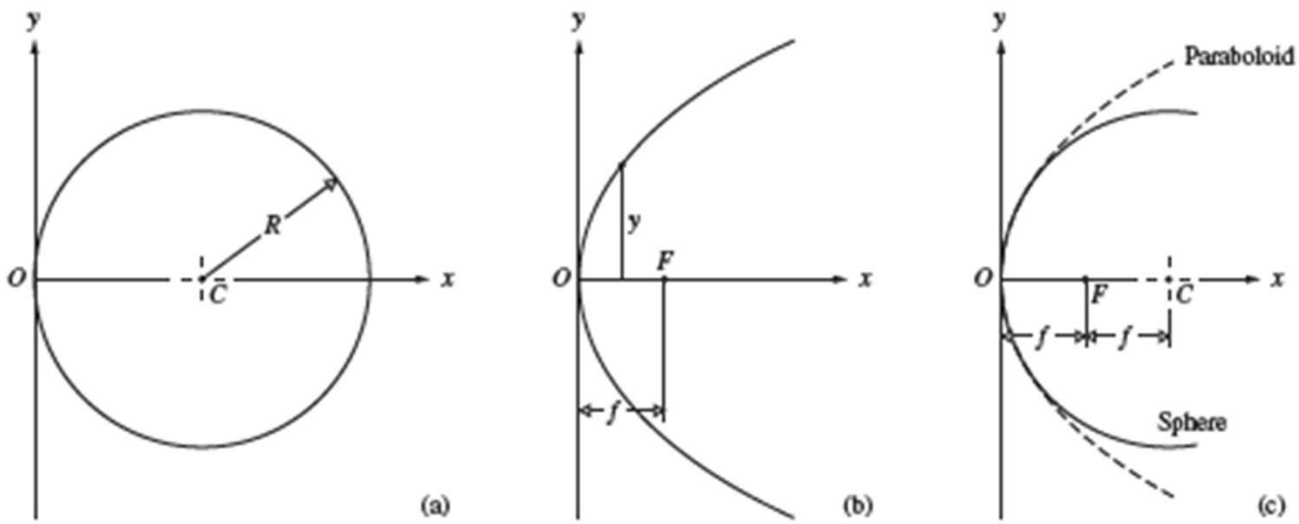
$$f = y + \frac{x(1 - \tan^2\theta)}{2\tan\theta}$$

b) Usando el resultado anterior, derive la expresión general para un espejo esférico de radio r. Además, aproxime para $\theta \ll 1$.

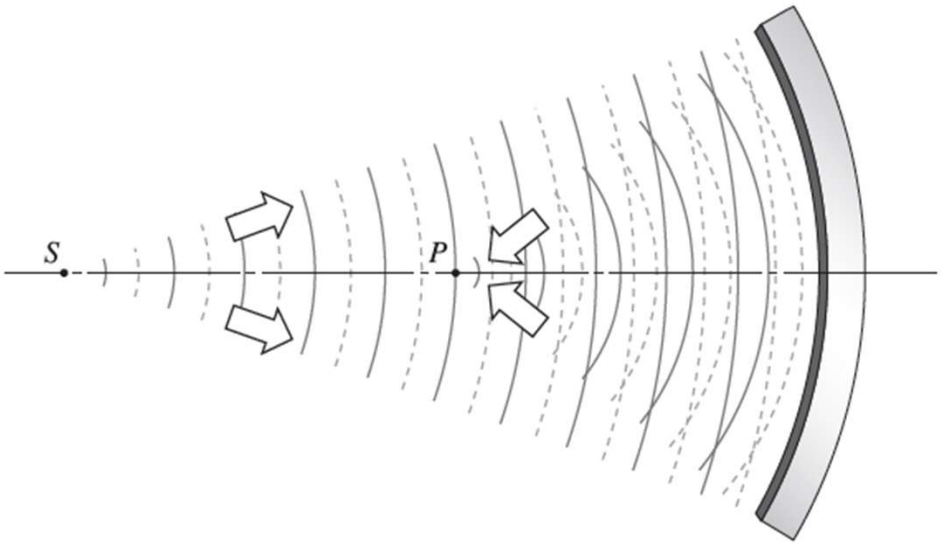


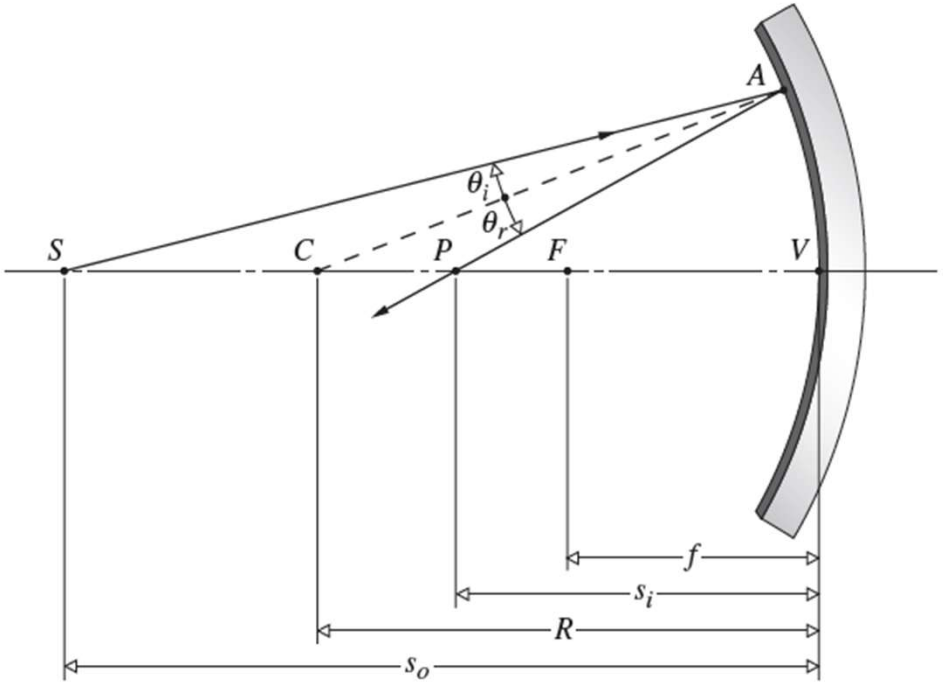
Espejos esféricos

En la **región paraxial** los espejos esféricos y paraboloides se comportan de igual forma.



Fórmula de espejos esféricos





Fórmula de espejos esféricos

$$\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}$$

TABLE 5.4 Sign Convention for Spherical Mirrors

Quantity	Sign	
	+	−
s_o	Left of V, real object	Right of V, virtual object
s_i	Left of V, real image	Right of V, virtual image
f	Concave mirror	Convex mirror
R	C right of V, convex	C left of V, concave
y_o	Above axis, erect object	Below axis, inverted object
y_i	Above axis, erect image	Below axis, inverted image